Detecting and Removing Gender Bias in Word Embeddings

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Abstract

When misused, machine learning may reveal human-induced data biases. Word embedding, a popular paradigm for collecting textual input as vectors for machine learning and natural language processing, causes this complexity. To that aim, we will present a way for altering an embedding to remove undesired connections like “nurse” and “female” while retaining others like ”queen” and “female”. We’ll concentrate on BERT, a transformer model that generates contextual word embeddings. Gender direction and BERT’s fill-in-the-mask task reveal gender prejudice. To debias the embedding, we will remove its component along the gender subspace and ensure that every gender-neutral term is separate from gender-specific word pairings.

Introduction

Word embeddings are word representations used in NLP to denote the representation of words for text analysis, but can also reveal biases. Recent research suggests that word embeddings may be skewed towards one gender. Several academic studies have examined word embeddings, but few have highlighted their sexism, particularly in contextualised ones. We propose a systematic and concise strategy to detect and reduce gender bias in word embeddings.

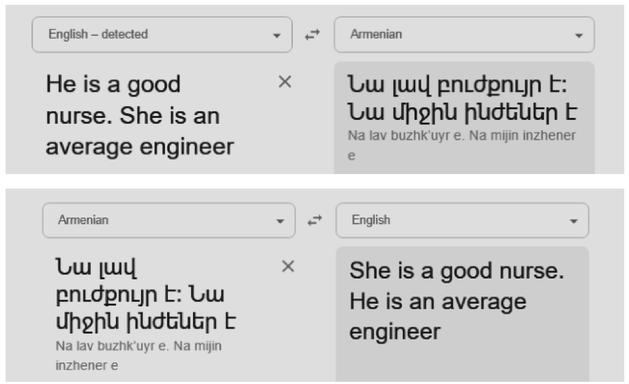


Figure 1. Gender Bias in Google Translate

Problem Statement

- Is there a gender bias in the contextual word embeddings?
- Can we work to mitigate the gender bias present in these embeddings?

Research Objective

- Analyze the existing bias for static and contextualized word embeddings.
- We look at strategies for identifying bias in contextualized word embeddings.
- We employ a BERT transformer model to find the existing bias.
- We continue to focus on eliminating the bias that the results we obtained demonstrate.
- We compare our model before and after debiasing.

Methodology

With the use of a multi-layer transformer architecture that includes both self-attention layers and feed-forward neural networks, BERT is able to process tokenized input text and generate contextualized word embeddings

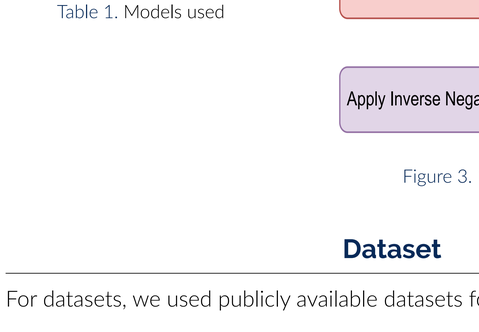


Figure 2. BERT architecture

Pretrained on two unsupervised task:

- Masked Language Modeling:
- Next Sentence Prediction

For detecting bias, we gave sentences to BERT with masked token to predict the word. Experiment was repeated with many variations of BERT.

Models
bert-base-uncased
bert-large-uncased
distilbert-base-uncased
roberta-base
roberta-large
albert-base-v1
albert-large-v1

Table 1. Models used

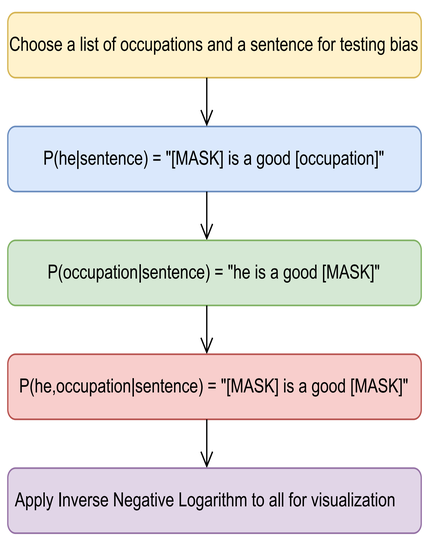


Figure 3. Mask Prediction Procedure

Dataset

For datasets, we used publicly available datasets for detecting gender bias.

- **Winogender Dataset:** Rudinger et al. (2018)’s [1] dataset includes sentences containing gender-neutral professions and gendered pronouns.
- **Pretrained on large dataset:** Wikipedia Dump and Book Corpus
- **Synthetic Dataset:** We created a synthetic dataset using the pretrained model instead of giving it data. We fed the model a few phrase structures and it created sentences which were used to identify gender bias.

Results and Analysis

- Masking the gendered pronoun:

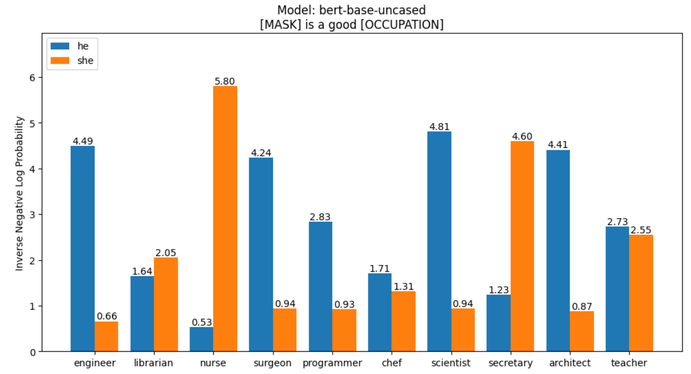


Figure 4. Predicting Gender Pronouns

- Masking the occupation:

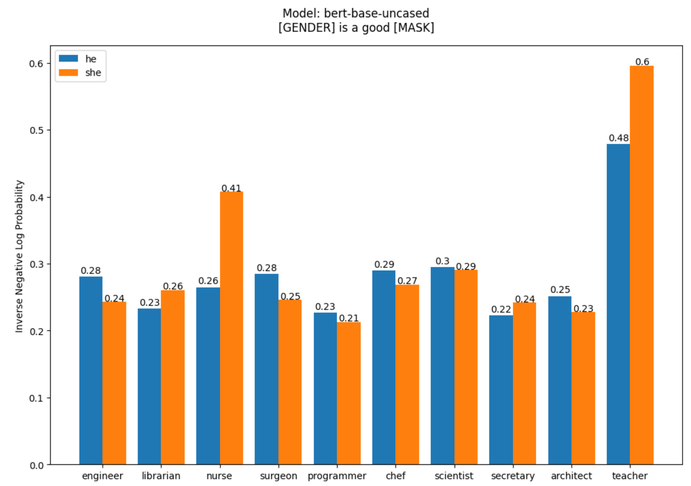


Figure 5. Predicting Occupations

- Masking both gendered pronouns and occupation:

Model	Input Sentence	Output Sentence
bert-base-uncased	[MASK] is a good [MASK]	he is a good teacher

Table 2. Predicting Gender Pronouns and Occupations

BERT is more likely to predict “he” than “she” in the case of engineer and “she” than “he” in the case of nurse. When asking BERT to assign a gender to a profession, it clearly demonstrates gender discrimination.

References

[1] Rachel Rudinger et al. “Gender Bias in Coreference Resolution”. In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. New Orleans, Louisiana: Association for Computational Linguistics, June 2018.

Github: https://github.com/AriyanH100/bias-in-word-embeddings